

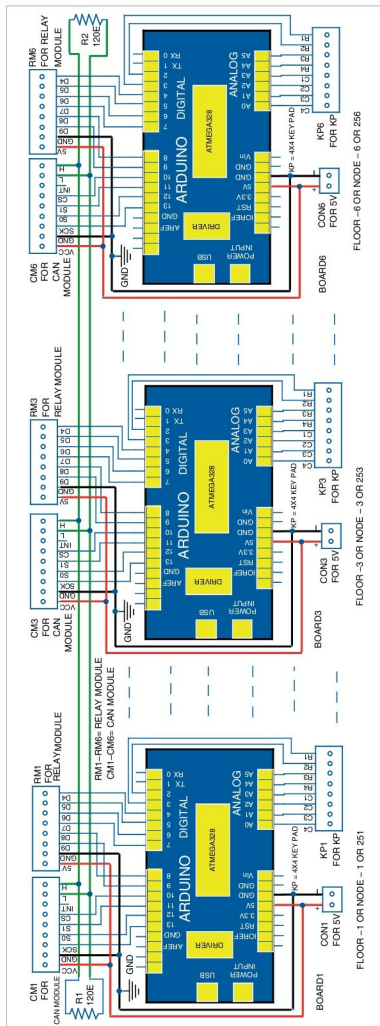


Fig. 2: Circuit diagram of CAN bus for a multistory building

the 5-digit switching signals on CAN bus node ID 253 will be 25341#. Here, the '#' key from the keypad is used as 'Enter' button for 'end of command' in the algorithm. To switch off the relay, you need to send 5-digit signal 25340# across the CAN bus.

Similarly, you can send 25391# and 25390# signals across the bus to switch on and switch off the sixth relay, respectively.

Assuming there is one CAN bus node on each floor in a six story building, this circuit will let you control up to six relays (and the devices connected to them) on any floor from any floor of the house. For example, if you wish to control all six relays on fourth floor that has CAN bus ID 254 (say, node 4), from the third floor having CAN bus ID 253 (node 3), you need to do the following:

First upload the Arduino code 'CAN_multistory253.ino' for third floor into Arduino Uno (Board3) and 'CAN_multistory254.ino' for fourth floor into Arduino Uno (Board4). When you enter 25341# keys from keypad (KP3), relay connected to D4 of Board3 energises and the load connected to its contacts switches on. When you enter 25340# keys from KP3, the relay de-energises and the load switches off.

Similarly, suppose you want to control the relays connected to node ID 254 from node ID 253. Enter 25441# from keypad (KP3); the relay connected to D4 pin of Arduino Uno (Board4) energises and the load connected to its contact switches on. Similarly, when you press 25440# from KP3, the relay de-energises.

CAN bus transceiver built on MCP2515 chip used in the project is shown in Fig. 4. It has an SPI bus male pin connector. Its 2-pin terminal connector is for connecting CAN bus—high (H) and low (L) cable wires. The cable should be of good quality shielded twisted pair copper type. A 120-ohm termination resistor at the end of the bus is essential, else it will weaken the signal through self-radiation.

Software

The software is written in Arduino programming language (Sketch) and compiled using Arduino IDE software. Software is just CAN_multistory.ino for all the nodes. Just include the node numbers like 253, 254, 255, 256, 257, 258, and 259 in the code, then upload the code to the corresponding Arduino nodes. This software is typically for an Arduino Uno-driven circuit.

As an example, we have used two Arduino source codes, CAN_multistory253.ino and CAN_multistory254.ino, for third and fourth floors, respectively. You can, like-wise, add the node numbers in the codes for the other Arduino Uno boards on respective floors.

Before compiling and uploading the code, you